Technical Assessment

The Smart Power & Load Orchestrator

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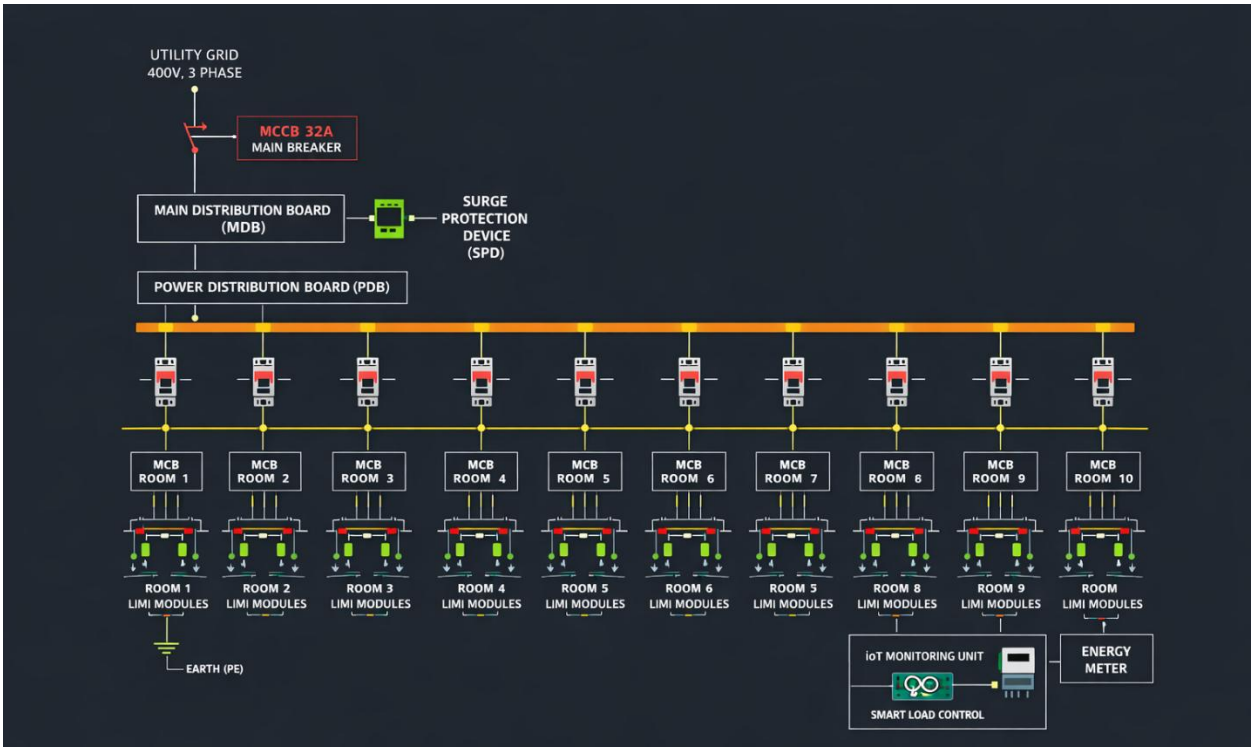
1. Electrical System Design (AutoCAD Electrical / ETAP)

1.1 Single-Line Diagram (SLD) - 10 Room Limi Smart Wing

The electrical distribution system of the Limi Smart Wing consists of a main incoming supply from the utility grid which feeds a Main Distribution Board (MDB). From the MDB, power is distributed to a Power Distribution Board (PDB) dedicated to the Limi modules installed in the 10 rooms.

Each room contains multiple smart modules such as ceiling lights, floor ambient lights, sensors, and control units.

1.2 Basic SLD Flow



Grid Supply → Main Circuit Breaker (MCCB) → Main Distribution Board →
Power Distribution Board → Individual Room Circuits → Limi Modules

1.3 Key Components:

- Incoming Supply: 3-Phase 400V AC
- Main Circuit Breaker: 63A MCCB
- Distribution Busbar: Copper Busbar System

- Outgoing Feeders: MCB protected circuits for each room
- Earthing System: Protective Earth connected to all metallic parts
- Surge Protection Device (SPD) at the MDB

This SLD ensures safe and reliable power delivery while allowing easy fault isolation.

1.4 Load Calculation

Assume the following specifications:

- Each Limi module = 60W
- Each room contains 5 modules

Total modules:

10 rooms × 5 modules = 50 modules

Total load:

50 × 60W = 3000W (3kW)

Adding 20% safety margin

Total design load:

3kW × 1.2 = 3.6kW

1.5 Current Calculation

Using the formula:

$$I = \frac{P}{V}$$

Assuming single-phase 230V supply:

$$I = \frac{3600}{230}$$

Current ≈ 15.65 A

Main Circuit Breaker

$$ICB=1.25\times Iload$$

ICB = 1.25 * 15.55 = 32A MCCB Selected

Room Circuit Breaker

$$Proom=300W$$

300/230 = 1.3 A

Applying safety margin:

$$I_{CB-room} \approx 2A$$

6A MCB preferred.

1.6 Circuit Breaker Selection

Component	Rating	Reason	Calculated Current
Main Breaker	32A MCCB	Handles full wing load	15.7A
Room Circuit Breaker	6A MCB	Each room loads 300W	1.3A
Control Circuit	2A MCB	Low-power IoT control	<1

Cable Selection:

- Main feeder: 6 mm² Copper
- Room circuits: 2.5 mm² Copper
- Control wiring: 1.5 mm²

1.7 Power Distribution Board (PDB) Layout

The PDB contains:

- Main MCCB

- Surge Protection Device (SPD)
- Busbars
- 10 outgoing MCB circuits (one per room)
- IoT monitoring module
- Energy meter

Protection hierarchy:

Grid → MCCB → Busbar → MCB → Load

1.8 Safety & Protection Scheme

Reflecting practices used in aviation infrastructure like Pakistan Civil Aviation Authority, the protection scheme includes multiple safety layers.

1.9 Relay Coordination

Protection devices are coordinated so that:

- MCB trips first for local faults
- MCCB trips only if downstream protection fails

This prevents system-wide outages.

1.10 Protection Devices

1. Overcurrent Protection
    - MCBs protect individual circuits.
2. Earth Leakage Protection
    - RCCB (30mA sensitivity) prevents electric shocks.
3. Surge Protection
    - SPD installed to protect smart electronics from lightning or switching surges.

4. Thermal Protection
    - Temperature sensors in the panel detect overheating.

2. IoT-Enabled Monitoring and Control (Arduino)

2.1 Smart Load Shedding System

The Limi Smart Wing includes a smart energy management system using Arduino Uno and current sensors.

2.2 Purpose:

Automatically reduce non-essential loads if power consumption exceeds a pre-set limit.

Threshold example: 5 kW

2.3 System Logic

i. Measure total current consumption using a current sensor.

ii. Calculate total power consumption.

iii. If power exceeds threshold:
    - Dim or switch off non-essential modules

iv. Maintain power to essential modules such as emergency lighting.

2.4 Arduino Logic (Example Code)

```
#define CURRENT_SENSOR A0
#define NON_ESSENTIAL_PIN 8
float threshold = 5000; // watts

void setup() {
  pinMode(NON_ESSENTIAL_PIN, OUTPUT);
  Serial.begin(9600);
}
```

```
void loop() {
  int sensorValue = analogRead(CURRENT_SENSOR);
  float current = sensorValue * 0.02;
  float voltage = 230;
  float power = voltage * current;

  Serial.println(power);

  if(power > threshold){
     digitalWrite(NON_ESSENTIAL_PIN, LOW); // dim lights
  }
  else{
     digitalWrite(NON_ESSENTIAL_PIN, HIGH);
  }

  delay(1000);
}
```

2.5 Sensor Inputs

Two types of sensors help monitor the system:

2.5.1 Current Sensor

Example: ACS712 Current Sensor Module

Function:

- Measures electrical current flowing through the circuit.
- Sends analog signal to Arduino.

This enables real-time power monitoring.

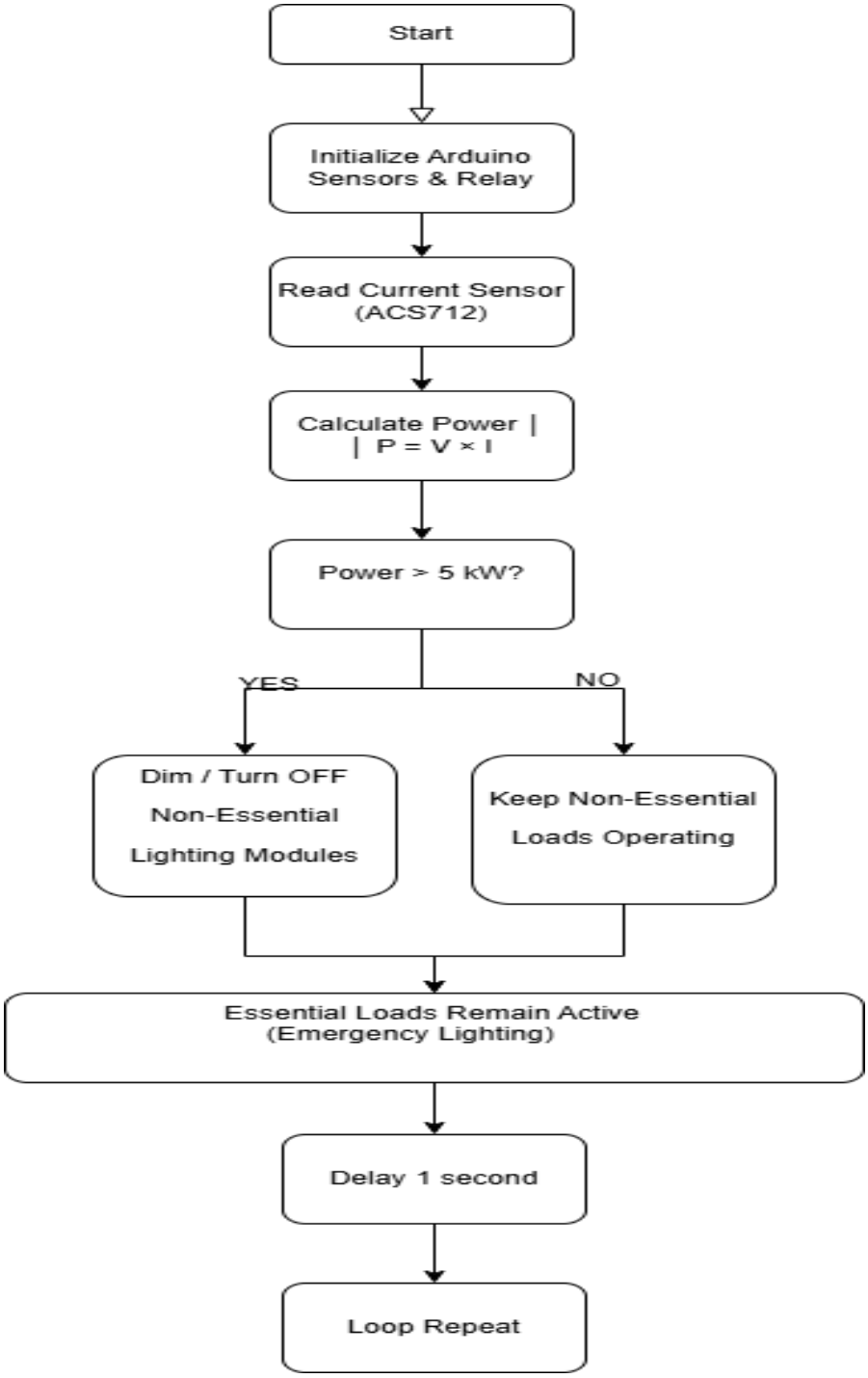
2.5.2 Light Sensor

Example: LDR Photoresistor Sensor

Function:

- Detects ambient light levels.
- Automatically reduces artificial lighting during daylight.

This further improves energy efficiency.



3. Renewable Energy Integration (Solar PV)

3.1 Solar Hybrid System

The Limi Smart Wing integrates a solar PV system to reduce dependence on grid electricity.

Key components:

- Solar Panels
- Hybrid Inverter
- Battery Storage
- Grid Connection

3.2 Dual-Axis Solar Tracking

Inspired by your project Dual-Axis Solar Tracker, the solar array adjusts its orientation to track the sun throughout the day.

Benefits:

- Up to 30–40% more energy generation
- Maximum sunlight exposure
- Improved energy yield

Tracking is done using:

- LDR sensors
- Servo motors
- Microcontroller control system

3.3 Solar System Capacity

Wing Load = 3.6 kW

Design Solar Capacity:

4 kW Solar PV System

Typical components:

- 10 × 400W panels
- Hybrid inverter
- Lithium battery storage

3.4 Energy Switching Logic

Power priority hierarchy:

1. Solar PV
2. Battery Storage
3. Grid Supply

3.5 Operation modes:

3.5.1 Daytime

Solar → powers Limi Wing directly

Excess energy → charges battery

3.5.1 Nighttime

Battery → supplies load

If battery low → grid supply activated

Automatic switching is handled by hybrid inverter.

3.6 Recommendations for Proposal

1. PV System Size: 8 kW
2. Hybrid Inverter: 10 kW, capable of automatic PV/Grid switching
3. Battery Storage: 10 kWh Li-ion (supports essential + partial non-essential load)
4. Dual-Axis Tracker: To maximize energy capture throughout the day
5. Load Shedding Logic: Non-essential loads dim when PV insufficient

4. Operations & HSE Management

4.1 Preventive Maintenance Checklist

Based on industrial maintenance experience like operations at PepsiCo bottling plants such as Sukkur Beverages (PepsiCo Bottler), preventive maintenance ensures reliability.

4.1.1 Monthly Inspection

- Check electrical connections
- Inspect cable insulation
- Verify circuit breaker operation
- Clean panel ventilation filters

4.1.2 Quarterly Inspection

- Thermal imaging of panels
- Test surge protection device
- Inspect solar panel alignment
- Check grounding resistance

4.1.3 Annual Inspection

- Complete system shutdown inspection
- Firmware updates for IoT modules
- Battery performance testing
- Calibration of sensors

4.2 HSE Risk Assessment - Ceiling Module Installation

Installing a Limi ceiling module requires technicians to work at height.

4.2.1 Identified Hazards

Hazard	Risk
Working at height	Falling from ladder
Electrical wiring	Electric shock
Dropped tools	Injury to personnel
Improper mounting	Ceiling collapse

4.3 Risk Control Measures

4.3.1 Fall Protection

- Use certified ladders
- Wear safety harness if above 2 meters

4.3.2 Electrical Safety

- Isolate circuit
- Lockout/Tagout procedure

4.3.3 PPE

- Helmet
- Insulated gloves
- Safety shoes
- Safety goggles

4.3.4 Tool Safety

- Use tethered tools
- Maintain clean work area

4.4 Environmental Considerations

- Proper disposal of electronic waste

- Using energy-efficient LED modules
- Reduce carbon footprint using solar power

4.5 Preventive Maintenance Schedule

Frequency	**Task**	**Responsible**	**Notes**
Daily	Visual inspection of modules	Technician	Check for visible damage
Weekly	Clean dust from panels & sensors	Technician	Ensure sensors work properly
Monthly	Check MCB operation, cable connections	Electrical Engineer	Thermal inspection optional
Quarterly	Test SPD, inspect battery, firmware sync	Engineer	Record all data in logbook
Annually	Full system inspection, calibration, software update	Project Manager	Ensure system reliability